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Preserving Multi-Modal Capabilities of Pre-trained VLMs for Improving Vision-Linguistic Compositionality

Youngtaek Oh[1] **Jae Won Cho**[2] **Dong-Jin Kim**[3] **In So Kweon**[1]* **Junmo Kim**[1]*

[1]KAIST [2]Sejong University [3]Hanyang University

[1]{youngtaek.oh, iskweon77, junmo.kim}@kaist.ac.kr
[2]chojw@sejong.ac.kr [3]djdkim@hanyang.ac.kr

Abstract

In this paper, we propose a new method to enhance compositional understanding in pre-trained vision and language models (VLMs) without sacrificing performance in zero-shot multi-modal tasks. Traditional fine-tuning approaches often improve compositional reasoning at the cost of degrading multi-modal capabilities, primarily due to the use of global hard negative (HN) loss, which contrasts global representations of images and texts. This global HN loss pushes HN texts that are highly similar to the original ones, damaging the model's multi-modal representations. To overcome this limitation, we propose Fine-grained Selective Calibrated CLIP (FSC-CLIP), which integrates local hard negative loss and selective calibrated regularization. These innovations provide fine-grained negative supervision while preserving the model's representational integrity. Our extensive evaluations across diverse benchmarks for both compositionality and multi-modal tasks show that FSC-CLIP not only achieves compositionality on par with state-of-the-art models but also retains strong multi-modal capabilities. Code is available at: https://github.com/ytaek-oh/fsc-clip.

1 Introduction

Humans naturally excel at multi-modal understanding, effortlessly perceiving and interpreting different modalities, such as images and text, and forming associations between them. This capability is evident in recognizing novel concepts (Fu et al., 2018), cross-modal retrieval (Kaur et al., 2021), and compositional reasoning (Levesque et al., 2012). To achieve this ability in artificial intelligence, foundational vision and language models (VLMs) have been trained on large-scale image-text datasets (Schuhmann et al., 2022b), significantly bridging the gap between human and machine capabilities in tasks like zero-shot recognition and image-text retrieval (Radford et al., 2021).

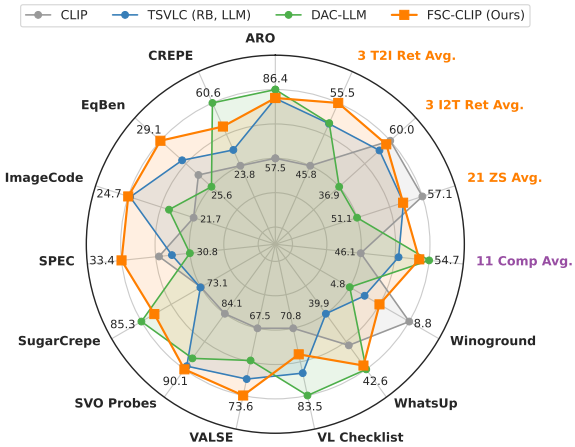

Figure 1: A holistic comparison of fine-tuning methods for vision-language compositionality. Enhancing compositionality often compromises multi-modal task performance in previous approaches. Our FSC-CLIP bridges this gap, minimizing these trade-offs. Full experimental results are provided in Tab. 1.

Despite these advances, VLMs still face challenges in compositional reasoning (Yuksekgonul et al., 2023). Humans intuitively understand complex compositional language in combination with images, engaging in spatial reasoning (Kamath et al., 2023b), recognizing attributes and relationships in objects (Hsieh et al., 2023), and perceiving equivariance between image and text (Wang et al., 2023). In contrast, VLMs often fail to understand these nuanced relationships (Liu et al., 2023a; Ray et al., 2023). This shortfall is attributed to their reliance on global, single vector representations (Kamath et al., 2023a) and limited ability to match compositional knowledge (Wang et al., 2024).

To improve compositionality in VLMs, both pre-training (Singh et al., 2023; Zheng et al., 2024) and fine-tuning (Zhang et al., 2024; Singh et al., 2024) methods have been proposed. In particular, fine-tuning, which leverages pre-trained knowledge and

*Corresponding authors

is cost-effective, is widely adopted in academia. Typically, this involves incorporating hard negative texts (Doveh et al., 2022, 2023; Herzig et al., 2023) into training. However, as shown in Fig. 1, this approach can result in a trade-off, where gains in compositionality come at the expense of performance in the multi-modal tasks: zero-shot classification (ZS) and image to text retrieval (I2T Ret). Previously, hard negative (HN) losses are applied to global image and text representations. Since HN texts are encoded too similarly to the original ones (Kamath et al., 2023a), pushing them away with the HN loss can disrupt the multi-modal representations.

To address this, we propose a new fine-tuning framework for VLMs that enhances compositional reasoning while preserving performance in multi-modal tasks. Our approach mitigates the degradation caused by global hard negative loss on single vector representations, which struggles to capture subtle informational differences between hard negative texts and the original text.

Our framework introduces two key innovations: **(1) Local Hard Negative (LHN) Loss**. We utilize dense alignments between image patches and text tokens to calculate the hard negative loss. This approach, inspired by the dense alignment for vision-language representation (Huang et al., 2021; Bica et al., 2024), aggregates local similarity scores to enhance compositional understanding without undermining multi-modal representations.
(2) Selective Calibrated Regularization (SCR). To address the adverse effects of hard negative (HN) losses caused by similarly encoded HN and original texts, SCR is designed to better regulate HN supervision. It selectively focuses on challenging HN texts and applies a slight positive margin, reducing confusion and improving calibration.

The whole framework, dubbed **F**ine-grained **S**elective **C**alibrated CLIP, offers fine-grained supervision of hard negatives while preserving the integrity of multi-modal representations. As shown in Fig. 1, FSC-CLIP not only improves compositionality but also maintains high performance in multi-modal tasks. It outperforms DAC-LLM in ZS and I2T Ret scores, while achieving similar compositionality (Comp) across a wide range of tasks. We summarize our contributions as follows:

- We propose a novel fine-tuning methodology, FSC-CLIP, that aims to enhance vision-language compositionality in pre-trained VLMs while maintaining their multi-modal task capabilities.
- We design a local hard negative (LHN) loss and a selective calibrated regularization (SCR) mechanism, effectively capturing subtle differences in hard negative texts and preserving the integrity of multi-modal representations.
- We validate FSC-CLIP through an extensive range of experiments, covering 11 compositionality, 21 zero-shot recognition, and 3 image-text retrieval tasks, establishing a comprehensive evaluation of VLMs' multifaceted capabilities.

2 Related Work

Contrastive Vision-Language Models. CLIP (Radford et al., 2021) has revolutionized multi-modal domains through large-scale image-text pre-training, demonstrating remarkable zero-shot capabilities. Its dual encoder architecture has introduced versatility and driven advancements across a wide range of existing vision (Zhou et al., 2022; Oh et al., 2022; Cho et al., 2022) and vision-language downstream tasks (Jang et al., 2022, 2023; Cho et al., 2023a,c,b; Kim et al., 2019, 2021a,b). CLIP also serves as the foundation for recognition (Liang et al., 2023), image captioning (Mokady et al., 2021; Lee et al., 2024; Kim et al., 2024a,b), large multi-modal models (Li et al., 2023; Liu et al., 2023b), and generative models (Podell et al., 2024). In addition, CLIP extends its utility to connecting 3D (Sun et al., 2024) or audio (Elizalde et al., 2023; Senocak et al., 2023) to language, highlighting its essential role in multi-modal and compositional tasks in practical applications. We aim to enhance CLIP's compositional understanding while preserving its multi-modal capabilities.
Vision-Language Compositionality. Although vision and language models exhibit promising capabilities such as zero-shot classification and retrieval (Radford et al., 2021; Zeng et al., 2022), they still struggle with compositional reasoning, which requires fine-grained understanding between image and text (Peng et al., 2024). Numerous benchmarks have been proposed, testing various aspects like attributes, relationships and objects (Zhao et al., 2022; Yuksekgonul et al., 2023), spatial reasoning (Kamath et al., 2023b; Liu et al., 2023a) and linguistic phenomena (Parcalabescu et al., 2022). To enhance compositionality, incorporating hard negative captions during fine-tuning has become a common approach (Zhang et al., 2024), with these captions being generated through rule-based methods (Doveh et al., 2022; Yuksekgonul et al.,

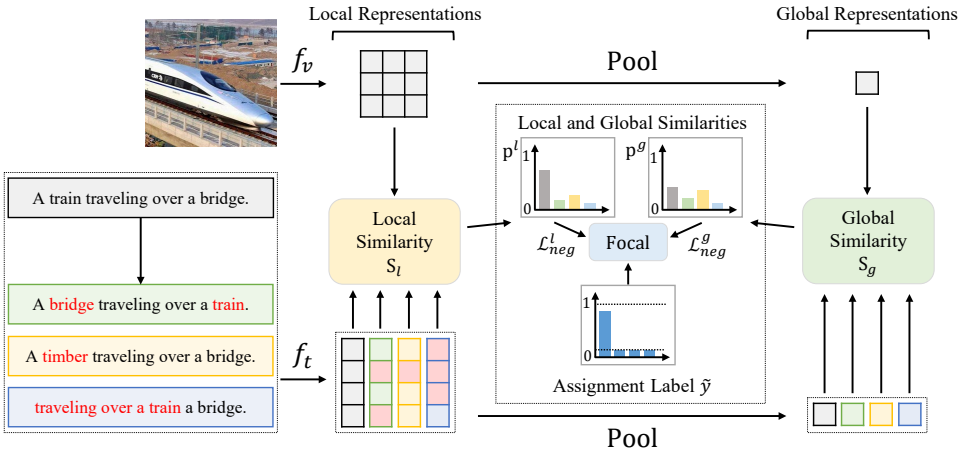

Figure 2: A complete FSC-CLIP framework that incorporates Local Hard Negative (LHN) Loss with Selective Calibrated Regularization (SCR), alongside a global HN loss. The LHN loss measures similarity between an image and a text at the patch and token levels to more accurately identify subtle differences between original and HN texts. SCR combines focal loss with label smoothing to mitigate the adverse effects of using hard negative losses.

2023), large language model prompting (Doveh et al., 2023), or scene graphs (Singh et al., 2023; Herzig et al., 2023). We comprehensively evaluate the capabilities of VLMs across a broad range of compositionality and multi-modal tasks.

3 Methodology

We first outline the fine-tuning setup for CLIP in Sec. 3.1. Next, we introduce FSC-CLIP, which incorporates **Local Hard Negative (LHN) Loss** and **Selective Calibrated Regularization (SCR)** in Secs. 3.2 and 3.3. The training objective for FSC-CLIP is described in Sec. 3.4. The complete FSC-CLIP framework, integrating both global and local HN losses with SCR, is illustrated in Fig. 2.

3.1 CLIP with Global Contrastive Loss

CLIP objective. Consider a mini-batch $\mathcal{B} = \{(I_i, T_i)\}_{i=1}^{B}$ of size B, consisting of image and text pairs (I_i, T_i). Using CLIP's visual and language encoders, $f_v(\cdot)$ (*e.g.*, ViT (Dosovitskiy et al., 2021)) and $f_t(\cdot)$ (*e.g.*, Transformers (Vaswani et al., 2017)), each image I_i is encoded into a sequence of visual tokens $\mathbf{V}_i = f_v(I_i)$, and each text T_i into a sequence of textual tokens $\mathbf{T}_i = f_t(T_i)$. These sequences are represented in a shared multi-modal space, with $\mathbf{V}_i = \{\mathrm{v}_{p,i}\}_{p=1}^{P}$ comprising P patch embeddings and $\mathbf{T}_i = \{\mathrm{t}_{w,i}\}_{w=1}^{W}$ consisting of W token embeddings. The global representations of image and text v_i and $t_i \in \mathbb{R}^d$ can be obtained by pooling the local representations: $v_i = \mathrm{Pool}(\mathbf{V}_i)$ and $t_i = \mathrm{Pool}(\mathbf{T}_i)$, respectively. For example, $\mathrm{Pool}(\cdot)$ corresponds to `avgpool` and `argmax` for images and texts in Radford et al. (2021).

CLIP aligns the corresponding images and texts by measuring the global-level similarity:

$$\mathrm{S}_g(I_i, T_i) = \exp\left(\cos\left(v_i, t_i\right)/\tau\right), \quad (1)$$

where $\cos(v, t) = \frac{v^T t}{\|v\| \cdot \|t\|}$. The image to text loss $\mathcal{L}_{i2t}$ of CLIP maximizes $\mathrm{S}_g(I_i, T_i)$, while minimizing $\mathrm{S}_g(I_i, T_j)$ for all non-matching texts $j \neq i$:

$$\mathcal{L}_{i2t} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\mathrm{S}_g(I_i, T_i)}{\sum_{j=1}^{B}\mathrm{S}_g(I_i, T_j)}, \quad (2)$$

and the text to image loss $\mathcal{L}_{t2i}$ is the reciprocal of $\mathcal{L}_{i2t}$ which aligns the matching image per text. The final CLIP loss is $\mathcal{L}_{\text{clip}} = \frac{1}{2}(\mathcal{L}_{i2t} + \mathcal{L}_{t2i})$.

Incorporating hard negative texts. To enhance the compositional reasoning of CLIP, hard negative (HN) texts are commonly incorporated into training, whether they are rule-based (Yuksekgonul et al., 2023) or generated by language models (Doveh et al., 2023). Consider a set of K different HN texts $\tilde{\mathrm{T}}_i = \{\tilde{T}_i^k\}_{k=1}^{K}$ originated from T_i. We introduce a separate hard negative loss added to $\mathcal{L}_{\text{clip}}$, similar to Doveh et al. (2022). First, we compute a similarity prediction probability p_i^g, assigned to the original caption T_i as follows:

$$p_i^g = \frac{\mathrm{S}_g(I_i, T_i)}{\mathrm{S}_g(I_i, T_i) + \sum_{k=1}^{K}\mathrm{S}_g\left(I_i, \tilde{T}_i^k\right)}. \quad (3)$$

Here, g represents the global representation, and the hard negative (HN) loss applied to this similarity assignment is formulated as cross entropy:

$$\mathcal{L}_{neg}^g = -\frac{1}{B}\sum_{i=1}^{B}\log p_i^g. \quad (4)$$

However, incorporating such global HN loss can inadvertently harm the multi-modal representations due to the similarly encoded global text representations between original and HN texts.

3.2 Local Hard Negative (LHN) Loss

To address this, we propose a novel Local Hard Negative (LHN) loss that utilizes a local similarity score $\mathrm{S}_l(I,T)$. Replacing the global similarity S_g with S_l, the LHN loss is formulated as follows:

$$\mathcal{L}_{neg}^{l} = \frac{-1}{B}\sum_{i=1}^{B}\log\underbrace{\frac{\mathrm{S}_l\left(I_i,T_i\right)}{\mathrm{S}_l\left(I_i,T_i\right)+\sum_{k=1}^{K}\mathrm{S}_l\left(I_i,\tilde{T}_i^k\right)}}_{p_i^l}, \tag{5}$$

where p_i^l represents the local similarity prediction.

Unlike Bica et al. (2024), which uses token-level contrast for image-text pairs, we introduce a new HN loss based on local similarity S_l from token-patch representations, enabling the capture of subtle differences between the original and HN texts.

Textual-aligned Visual Patches. $\mathrm{S}_l(I,T)$ is designed to measure the similarity between token and patch embeddings for each token in the given text T. From the patch representations $\mathbf{V}=\{\mathrm{v}_p\}_{p=1}^{P}$, we first derive the textual-aligned patch embeddings $\hat{\mathbf{V}}=\{\hat{\mathrm{v}}_w\}_{w=1}^{W}$, corresponding to each textual token feature t_w in $\mathbf{T}\in\mathbb{R}^{W\times d}$. This is achieved by performing a weighted average of patches $\mathbf{V}$ using attention weights $\mathrm{a}\in\mathbb{R}^{W\times P}$ derived from normalizing the similarity map s between token and patch embeddings. We denote the similarity map as $\mathrm{s}=\mathbf{T}^T\mathbf{V}\in\mathbb{R}^{W\times P}$, where $\mathrm{s}_{w,p}=\mathrm{t}_w^T\mathrm{v}_p$.

To relate multiple similar patches for each token, we min-max normalize s to obtain a:

$$\mathrm{a}_{w,p}=\frac{\mathrm{s}_{w,p}-\min_k \mathrm{s}_{w,k}}{\max_k \mathrm{s}_{w,k}-\min_k \mathrm{s}_{w,k}}, \tag{6}$$

and use the attention weights a to aggregate $\mathbf{V}$, obtaining the textual-aligned patches $\hat{\mathbf{V}}=\{\hat{\mathrm{v}}_w\}_{w=1}^{W}$:

$$\hat{\mathrm{v}}_w=\frac{1}{\sum_{p=1}^{P}\mathrm{a}_{w,p}}\cdot\sum_{p=1}^{P}\mathrm{a}_{w,p}\cdot\mathrm{v}_p. \tag{7}$$

In Appendix B.1, we explore different normalization choices for the attention weights in Eq. (6).

Token-level Similarity. After obtaining the textual-aligned visual tokens $\hat{\mathbf{V}}$, we aggregate the per-token similarities between $\hat{\mathbf{V}}$ and $\mathbf{T}$ as follows:

$$\mathrm{S}_l\left(I,T\right)=\sum_{w=1}^{W}\exp\left(\cos\left(\hat{\mathrm{v}}_w,\mathrm{t}_w\right)/\tau\right), \tag{8}$$

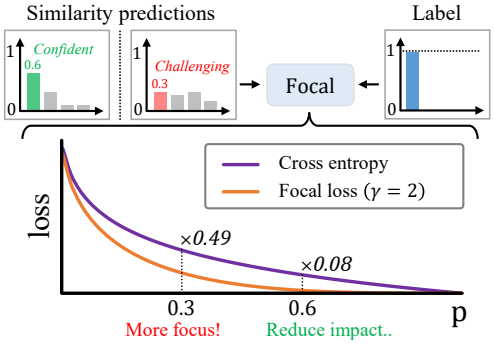

Figure 3: A conceptual illustration of the confidence-based weighting mechanism in HN loss. It reduces the adverse impact of HN supervision by lowering the signal from confident predictions while selectively focusing on challenging ones, crucial for learning compositionality.

where $\hat{\mathrm{v}}_w\in\hat{\mathbf{V}}$ and $\mathrm{t}_w\in\mathbf{T}$. Unlike $\mathrm{S}_g(I,T)$ which is based on global representations, $\mathrm{S}_l(I,T)$ focuses on the local alignment between image and text, better distinguishing features between correct and HN texts, thereby reducing the negative impact by the hard negative loss, as illustrated in Fig. 2.

We observe that $\mathcal{L}_{neg}^{l}$ maintains multi-modal task performance close to the pre-trained representations while significantly enhancing compositionality. Notably, the order of aggregation, whether pooling first and then computing similarity (*e.g.*, S_g), or computing token-level similarity before aggregation (*e.g.*, S_l), proves to be important.

3.3 Selective Calibrated Regularization (SCR)

Since hard negative (HN) texts are often encoded similarly to the original texts, HN losses can disrupt multi-modal representations. To counter this, we propose Selective Calibrated Regularization (SCR) to better regulate HN supervision, seamlessly applicable to both global and local HN losses.

SCR has two components: one modulates the supervision signal based on image-text similarity, while the other adjusts label assignments to calibrate the positiveness of HN texts. As shown in Tab. 2, we confirm that both components are crucial for preserving the representation integrity.

Focal Loss to Target Challenging HN Texts. To mitigate the negative impact of supervising HN texts, we reduce the supervision signal for confident similarity predictions to the original text. Instead, we focus more on challenging HN texts that exhibit higher similarity to the image and may be confused with the original texts. This confidence-based weighting aligns with the concept of focal loss (Lin et al., 2017), as shown in Fig. 3.

Formally, let the similarity prediction for the i-th batch item, including K generated HN texts, be represented as a vector $\mathrm{p}_i \in \mathbb{R}^{1+K}$, where the first element corresponds to the original text. The HN loss can be re-formulated in a vector representation with p_i as $\mathrm{CE}(\mathrm{p}_i, \mathrm{y}_i) = \sum_{k=0}^{K} l_{i,k}$, where $l_{i,k} = -\mathrm{y}_{i,k} \log \mathrm{p}_{i,k}$ and $\mathrm{y}_i = \mathbb{1}_{[k=0]} \in \mathbb{R}^{1+K}$ indicates the assignment label between an image and all texts. To reduce the negative impact of the confident image-text similarity predictions, we apply confidence-based weighting to CE loss as follows:

$$\text{Focal}\left(\mathrm{p}_i, \mathrm{y}_i\right) = \sum_{k=0}^{K} \left(1 - \mathrm{p}_{i,k}\right)^{\gamma} l_{i,k}, \qquad (9)$$

where γ is the modulation parameter. This strategy prioritizes challenging image-text associations, essential for learning compositionality, while effectively preventing degradation from the HN loss.

Label Smoothing to Calibrate the Positiveness of HN Texts. Although hard negative (HN) texts share similar representations with the original text, previous methods have overlooked their potential positiveness in the HN loss design, assigning a strict value of 0 to all HN texts in the label vector y_i. Similar to the motivation in Zhang et al. (2024), but differing from their ranking loss approach, we acknowledge the potential correctness of HN texts by assigning a slight positive margin rather than categorizing them as entirely negative.

To this end, we apply label smoothing (Guo et al., 2017) to the label vector y_i using a smoothing parameter β to ensure a positive margin for HN texts:

$$\tilde{\mathrm{y}}_{i,k} = (1 - \beta) \cdot \mathrm{y}_{i,k} + \frac{\beta}{1 + K}, \qquad (10)$$

where $\tilde{\mathrm{y}}_i$ provides a non-binary label for the HN losses. It helps preserve the model's representations during training with HN losses.

3.4 Overall Training Objective

Our `FSC-CLIP` incorporates two hard negative (HN) losses, $\mathcal{L}^{g}_{neg}$ and $\mathcal{L}^{l}_{neg}$, representing global and local HN losses respectively, into CLIP loss $\mathcal{L}_{\text{clip}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{clip}} + \lambda_g \mathcal{L}^{g}_{neg} + \lambda_l \mathcal{L}^{l}_{neg}, \qquad (11)$$

where λ_g and λ_l are the weighting factors for the respective losses. Selective Calibrated Regularization (SCR) is applied to both losses, incorporating label smoothing and focal loss. The global HN loss, $\mathcal{L}^{g}_{neg}$ is computed as Focal $(\mathrm{p}^g, \tilde{\mathrm{y}})$, while the LHN loss, $\mathcal{L}^{l}_{neg}$ is derived similarly, by replacing p^g with p^l for the local representations.

4 Experiments

Training Datasets. We consider three image-text datasets for fine-tuning: COCO captions (Chen et al., 2015), CC-3M (Sharma et al., 2018), and LAION-COCO (Schuhmann et al., 2022a). For COCO captions, we utilize 100K examples preprocessed by Yuksekgonul et al. (2023). As pointed out by Singh et al. (2023), COCO shares data with several evaluation benchmarks (*e.g.*, SugarCrepe and retrieval tasks), which may inadvertently affect the results. To ensure a broader evaluation and avoid such overlap, we additionally consider CC-3M and LAION-COCO for fine-tuning. For each dataset, we randomly sample 100K examples and, instead of using raw captions, we utilize synthetic captions paired with images. Specifically, for CC-3M, we generate captions using CoCa (Yu et al., 2022) with ViT-L/14, while for LAION-COCO, we use captions generated by BLIP (Li et al., 2022b) with ViT-L/14, applied to the LAION-2B dataset (Schuhmann et al., 2022b).

Hard Negative (HN) Texts. We employ simple rule-based methods for generating hard negative (HN) texts, avoiding the need for external language models like Le Scao et al. (2023) used in Doveh et al. (2023). For each original caption, we apply three distinct operations: `negclip`, `replace`, and `bi-gram shuffle`. These operations are applied at every training step, ensuring variation in HN texts across iterations. As a result, each batch item is paired with an image and four captions, as illustrated in Fig. 2. Further details and examples on these operations are provided in Appendix A.1.

Training Setup. Consistent with previous methods (Yuksekgonul et al., 2023; Singh et al., 2023; Zhang et al., 2024), we trained our models during 5 epochs with batch size 256, using OpenCLIP repository (Ilharco et al., 2021). The learning rate is set to 5e-6 and decayed by a cosine schedule, with a warmup of 50 steps. Models are optimized using AdamW with a weight decay of 0.1. We use a single Quadro RTX 8000 GPU with 48GB memory for training. Images are re-scaled to 224, and the context length is 77 for texts. We set the weighting factors $\lambda_g = 0.5$ and $\lambda_l = 0.2$. For SCR, we set $\gamma = 2.0$ and $\beta = 0.02$ for focal loss and label smoothing, respectively. We also experiment with LoRA (Hu et al., 2022), which preserves the original model parameters. Consistent with Doveh et al. (2022, 2023), we set the rank to 4. Training our model takes less than one hour for 100K samples.

Method	LoRA	ARO	CREPE	EqBen	ImageCoDe	SugarCrepe	SVO Probes	VALSE	VL-Checklist	WhatsUp	Winoground	SPEC	Comp	ZS	I2T Ret	T2I Ret
CLIP (ViT-B/32)		57.5	23.8	26.5	21.7	73.1	84.1	67.5	70.8	41.5	8.8	31.9	46.1	57.1	60.0	45.8
Fine-tuned: MS-COCO, 100K Samples																
NegCLIP[1]		80.9	30.3	**30.3**	26.4	83.7	90.8	73.7	74.9	**42.9**	8.0	**34.6**	52.4	55.9	66.8	58.4
CE-CLIP[2]		76.3	34.7	26.8	24.5	**85.7**	90.1	**76.7**	76.9	41.7	5.2	33.0	52.0	49.9	59.2	57.4
GNM-CLIP[3]		57.1	17.4	28.3	25.0	78.7	89.2	71.1	70.6	42.1	**10.2**	33.1	47.5	**56.3**	66.1	55.5
MosaiCLIP[†,4]		82.6	-	-	-	-	90.7	-	76.8	-	-	-	-	-	-	-
NegCLIP[‡]		85.0	34.7	29.8	26.2	84.5	90.6	74.7	75.4	41.2	8.2	34.2	53.1	55.1	66.1	57.9
FSC-CLIP (Ours)		85.1	42.2	29.8	26.3	85.1	**90.9**	75.3	76.7	40.6	9.5	34.2	**54.2**	55.7	66.3	58.3
FSC-CLIP (Ours)	✓	**85.2**	**42.9**	29.7	**26.5**	82.1	90.4	75.0	**77.2**	41.7	6.0	33.2	53.6	55.6	65.3	57.2
*Fine-tuned: Conceptual Captions - 3M (CC-3M), **3M** Samples*																
TSVLC[5] (RB)	✓	83.5	36.1	27.4	24.0	76.9	**89.8**	69.3	77.5	40.9	6.8	31.6	51.2	54.9	54.9	52.1
TSVLC[5] (RB+LLM)	✓	82.7	33.1	27.6	24.6	73.2	89.7	72.2	79.2	39.9	5.8	31.4	50.9	**55.4**	55.1	52.3
DAC-LLM[6]	✓	86.4	60.6	25.6	22.8	**85.3**	88.9	70.5	83.5	42.6	4.8	30.8	54.7	51.1	36.9	52.4
DAC-SAM[6]	✓	83.3	**63.7**	25.3	24.3	83.8	88.5	70.2	**84.7**	42.4	**8.5**	29.9	**55.0**	51.9	41.1	49.0
MosaiCLIP[†,4]		80.4	-	-	-	-	-	-	77.3	-	-	-	-	53.5	-	-
*Fine-tuned: Conceptual Captions – 3M (CC-3M), **100K** Samples*																
NegCLIP[‡]		**86.5**	50.5	25.8	24.6	83.4	88.6	72.4	79.0	**43.2**	7.0	32.8	54.0	52.6	51.8	54.1
FSC-CLIP (Ours)		78.8	44.0	**28.5**	**25.2**	84.3	88.2	**74.9**	77.4	42.6	6.8	**34.2**	53.2	53.5	55.8	**54.6**
FSC-CLIP (Ours)	✓	84.4	50.6	27.7	24.5	82.3	88.8	74.5	80.3	42.1	5.0	32.2	53.9	53.6	**56.1**	54.0
*Fine-tuned: LAION-COCO, **600M** Samples*																
CLoVe[7]		83.0	41.7	26.9	**25.3**	**84.6**	87.9	71.8	66.6	41.8	6.5	31.7	51.6	51.0	53.1	**56.0**
*Fine-tuned: LAION-COCO, **100K** Samples*																
NegCLIP[‡]		**86.4**	48.7	27.2	**25.3**	80.9	89.6	70.9	76.0	**43.0**	**7.8**	32.3	53.5	54.1	52.3	54.1
FSC-CLIP (Ours)		82.8	46.8	**29.1**	24.7	82.6	**90.1**	**73.6**	75.7	42.4	6.8	**33.4**	53.5	55.3	**58.2**	55.5
FSC-CLIP (Ours)	✓	85.5	**54.4**	**29.1**	24.9	80.6	89.7	72.6	**78.4**	42.8	5.8	32.5	**54.2**	**55.9**	57.3	54.3

[†]Numbers taken from the original paper. [‡]Our implementation, without additional image batch.

References: [1](Yuksekgonul et al., 2023) [2](Zhang et al., 2024) [3](Sahin et al., 2024) [4](Singh et al., 2023) [5,6](Doveh et al., 2022, 2023) [7](Castro et al., 2024)

Table 1: A holistic comparison of fine-tuning methods applied to the pre-trained CLIP ViT-B/32 model across 11 compositionality, 21 zero-shot classification, and 3 retrieval tasks, including their meta averages: `Comp`, `ZS`, and `I2T/T2I Ret`. `FSC-CLIP` achieves superior compositionality scores while maintaining strong multi-modal task performances. For each fine-tuning dataset, the best numbers are **bold**, and the second-best numbers are underlined.

Evaluation Setup. We utilize an *extensive* range of benchmarks for a comprehensive evaluation, exceeding the scope of previous works. Full details including references are provided in Appendix A.2.

For compositionality, we employ 11 benchmarks in total: ARO, CREPE-Productivity, EqBen, ImageCoDe, SPEC, SugarCrepe, SVO Probes, VALSE, VL-Checklist, WhatsUp, and Winoground, testing different facets of compositional reasoning. For multi-modal tasks, we evaluate 21 zero-shot classification tasks using ELEVATER toolkit. In addition, we conduct image-text retrieval evaluations on COCO, Flickr30k, and COCO-Counterfactuals. All those evaluations are performed using the `vl-compo` package (Oh et al., 2024).

We report a single aggregated number, which is the average of sub-tasks for each compositionality benchmark. We also provide the meta-average across all compositionality benchmarks (`Comp`), the average performance over 21 zero-shot classification tasks (`ZS`), and the average Recall@1 for three image to text (`I2T Ret`) and text to image (`T2I Ret`) retrieval tasks, as shown in Tab. 1. For a fair comparison, we consistently run evaluations for all the previous models across all the benchmarks.

4.1 Main Results

We compare our `FSC-CLIP` to previous fine-tuning methods for compositionality. We report both compositionality and multi-modal task performance as shown in Tab. 1. In Fig. 4, we visualize the trade-off trajectory between `Comp` and `ZS` through the robust fine-tuning method (Wortsman et al., 2022).

Compositionality while Sacrificing Multi-Modal Tasks. We introduce our baseline, NegCLIP[‡], serving as a direct comparison to our `FSC-CLIP`. Unlike the original implementation of NegCLIP (Yuksekgonul et al., 2023), we utilize an online version of hard negatives generation (*e.g.*, `negclip`) and omit the use of additional similar image batches. This baseline will be further used in our ablation study, with the symbol [‡] omitted for convenience.

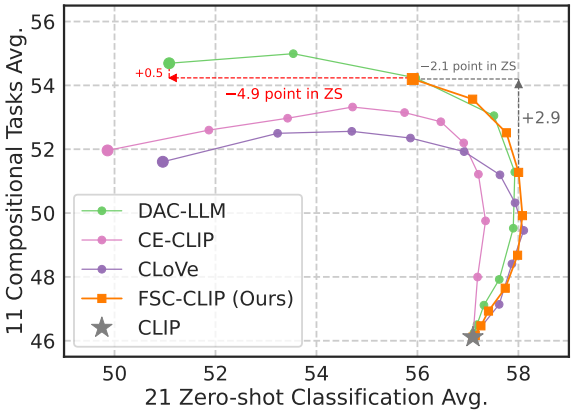

Figure 4: Fine-tuning trajectories between compositionality (Comp) and zero-shot classification (ZS) via robust fine-tuning method (Wortsman et al., 2022). Each point represents the interpolated model between the pre-trained and each fine-tuned version, at varying ratios. FSC-CLIP offers better trade-offs between Comp and ZS, maintaining ZS scores in the fully fine-tuned model.

As shown in Tab. 1, we first compare our FSC-CLIP with previous models fine-tuned on COCO, aligning our results with those in the literature. CE-CLIP[2] shows a significant drop in ZS score to 49.9. Meanwhile, GNM-CLIP[3] maintains a ZS score close to that of the pre-trained model, but shows only a modest increase in Comp. In contrast, our model achieves superior Comp scores while maintaining competitive ZS and retrieval performance. As note, we have grayed out the retrieval scores of models fine-tuned on COCO to account for the influence of overlapping data.

When fine-tuned on datasets other than COCO, such as CC-3M and LAION-COCO, all baseline models show improvements in the Comp score, but this comes at the expense of their ZS and I2T Ret scores compared to the pre-trained CLIP. For example, NegCLIP[‡] demonstrates promising Comp scores compared to methods like TSVLC[5] and CLoVe[7], but still shows weaker ZS and I2T Ret scores relative to the pre-trained model. Similarly, DAC-LLM[6], despite having the strongest Comp score supported by LLM-augmented captions, suffers notable declines in both ZS and I2T Ret, decreasing by 6.0 and 23.1 points, respectively. Although TSVLC[5] preserves these scores better than other models, its Comp score improvements are relatively smaller. These methods apply hard negative (HN) loss to global-level representations, potentially causing the observed performance drops.

Preserving Multi-Modal Tasks. FSC-CLIP stands out by achieving higher Comp scores than previous models, comparable to DAC-LLM, while maintaining strong performance in multi-modal tasks. As shown in Fig. 1, when fine-tuned on a 100K subset of LAION-COCO, our model achieves a Comp score of 53.5, significantly surpassing its pre-trained counterpart, and a ZS score of 55.9, nearly matching the pre-trained CLIP. Additionally, it attains an I2T Ret score of 58.2, the highest among models not fine-tuned on COCO. Further improvements are observed with using LoRA (Hu et al., 2022) for fine-tuning, which boosts the Comp score to 54.2 while maintaining the ZS score. Similar trends are evident when we fine-tune FSC-CLIP on a 100K subset of CC3M. Remarkably, these results are achieved by our innovative Local HN loss and Selective Calibrated Regularization (SCR) design. We further analyze these contributions in Sec. 4.2.

id	$\mathcal{L}^g_{neg}$	$\mathcal{L}^l_{neg}$	Focal	LS	Comp	ZS	I2T Ret	T2I Ret
1	✓	-	-	-	54.0	53.6	47.4	53.7
2	-	✓	-	-	51.7	55.7	61.6	54.5
3	✓	✓	-	-	54.4	52.6	46.9	53.8
4	✓	✓	✓	-	54.2	54.2	53.1	54.8
5	✓	✓	-	✓	53.9	53.8	51.7	54.9
6	✓	✓	✓	✓	53.5	55.3	58.2	55.5
7	✓	-	✓	✓	52.8	55.3	57.1	55.6
8	-	✓	✓	✓	50.2	55.9	63.2	55.1

Table 2: Impact by individual component. The local HN loss preserves multi-modal task performance. In addition, focal loss and label smoothing (LS) in SCR complement each other, improving the decreased multi-modal task performance caused by the HN losses.

Robust Fine-tuning on Compositionality and Zero-shot Tasks. As depicted in Fig. 4, we utilize the weight-space ensembling technique, WiSE-FT (Wortsman et al., 2022), to compare different fine-tuning methods and their trajectories, specifically in terms of Comp and ZS scores using LAION-COCO for fine-tuning our model. We create intermediate models by interpolating between each fine-tuned model and the pre-trained one. The blending ratio increases from 0.0 (*e.g.*, pre-trained) to 1.0 (*e.g.*, fully fine-tuned), in increments of 0.1.

FSC-CLIP with LoRA attains a ZS score of 58 at the intermediate, surpassing the scores of other models, while improving Comp to 50. When fully fine-tuned, it attains superior Comp score and offers better trade-offs than CLoVe and CE-CLIP, without significant loss in ZS. In contrast, DAC-LLM sees a significant drop in ZS, gaining only 0.5 point in Comp, as highlighted by the red marker. Meanwhile, FSC-CLIP not only matches but exceeds the ZS score by 4.9 in the fully fine-tuned model.

id	λ_l	Comp	ZS	I2T Ret	T2I Ret
1	-	52.9	55.8	57.5	55.5
2	0.1	53.0	55.7	57.4	55.4
3	0.2	53.5	55.3	58.2	55.5
4	0.5	53.5	55.7	57.3	55.4

(a) Sensitivity to the weighting factor λ_l of the local HN loss.

id	γ	Comp	ZS	I2T Ret	T2I Ret
1	-	53.9	53.8	51.7	54.9
2	1.0	53.4	54.9	54.7	55.1
3	2.0	53.5	55.3	58.2	55.5
4	5.0	52.3	55.6	60.2	55.5

(b) Sensitivity to the modulation factor γ of focal loss.

id	β	Comp	ZS	I2T Ret	T2I Ret
1	-	54.2	54.2	53.1	54.8
2	0.02	53.5	55.3	58.2	55.5
3	0.05	53.1	55.2	59.0	55.1
4	0.10	52.3	55.2	58.7	55.3

(c) Sensitivity to the label smoothing factor β.

Table 3: Sensitivity analysis of each component in our FSC-CLIP framework. **(a):** With the global HN loss applied, applying the local HN loss benefits the compositionality while preserving the multi-modal task scores. **(b)** and **(c)**: Both focal loss and label smoothing, the two components of our Selective Calibrated Regularization (SCR), mutually enhance multi-modal task performance but may compromise compositionality when applied too strongly. We highlight the cells corresponding to our design choices in the final FSC-CLIP model.

CLIP[1]	LoRA	Comp	ZS	I2T Ret	T2I Ret
ViT-B/16		46.2	60.3	62.9	49.0
+ NegCLIP		54.1	55.9	53.8	58.1
+ **FSC-CLIP**		54.1	57.0	59.7	59.3
+ **FSC-CLIP**	✓	54.6	57.4	59.9	58.8

[1]Pre-trained: 400M OpenAI, Fine-tuned: LAION-COCO 100K subset.

Table 4: Fine-tuning results of CLIP with a ViT-B/16 encoder, pre-trained on 400M samples of OpenAI data.

CLIP[2]	LoRA	Comp	ZS	I2T Ret	T2I Ret
ViT-B/32		44.3	63.0	63.8	51.2
+ NegCLIP		53.5	59.2	52.1	52.3
+ **FSC-CLIP**		52.9	61.1	56.8	53.8
+ **FSC-CLIP**	✓	54.0	60.7	56.8	53.1

[2]Pre-trained: DataComp-XL, Fine-tuned: LAION-COCO 100K subset.

Table 5: Fine-tuning results of CLIP with a ViT-B/32 encoder, pre-trained on 12.8B DataComp-XL.

4.2 Analysis

We further present an in-depth analysis on our FSC-CLIP, fine-tuned on LAION-COCO:

Impact of Individual Components. From Tab. 2, we observe that applying the local HN loss alone (row 2) surprisingly preserves the multi-modal scores. However, when both global and local HN losses are activated (row 3), Comp is further boosted but at the cost of ZS and I2T Ret scores, likely due to the complicated adverse effects of the losses. The proposed SCR effectively addresses this degradation. Both focal loss (row 4) and label smoothing (row 5) are effective and, when combined, complementarily boost all the ZS, I2T Ret, and T2I Ret scores. Notably, I2T Ret increases by 11.3 (rows 3 to 6) with only a relatively mild drop in Comp. We also note that comparing rows 7 and 8 with rows 1 and 2, SCR significantly boosts multi-modal task scores. Furthermore, as shown in row 6, applying both global and local HN losses is essential for achieving better Comp and I2T Ret scores.

Sensitivity Analysis. We explore the impact of individually varying each component's parameters in the final model, as detailed in Tab. 3. From Tab. 3a, we find that increasing the local HN loss parameter λ_l improves Comp score while preserving multi-modal task scores. Tab. 3b shows that increasing the modulation parameter γ boosts multi-modal tasks; however, beyond a certain point, we find that compositionality declines, as weakening the learning signal from HN texts. Similarly, Tab. 3c indicates that label smoothing benefits multi-modal tasks, particularly I2T Ret. Yet, assigning too much positive margin with β to negative samples can impede the learning of compositionality.

Fine-tuning CLIP with ViT-B/16. We also fine-tuned CLIP with a ViT-B/16 encoder from OpenAI for comparison, as detailed in Tab. 4. This model uses more image patches in training, showing better multi-modal capabilities. However, no gains are observed in Comp compared to the ViT-B/32 model from Tab. 1. After fine-tuning, NegCLIP decreases ZS and I2T Ret scores. In contrast, FSC-CLIP maintains its Comp score and significantly enhances multi-modal task performances. We also find that fine-tuning with LoRA yields improved ZS and I2T Ret scores, along with a higher Comp score.

Scaling Pre-training Data for Fine-tuning. We explore the effect of large-scale pre-training data when fine-tuned. From Tab. 5, we fine-tuned a CLIP model with a ViT-B/32 encoder, pre-trained on 12.8B DataComp-XL dataset (Gadre et al., 2023), far exceeding the 400M samples from OpenAI (Radford et al., 2021). Despite the larger scale pre-training yielding a promising ZS score of 63.0, we find no improvement in compositionality compared to OpenAI's CLIP. For fine-tuning, NegCLIP results in a notable drop in multi-modal task performance. In contrast, FSC-CLIP with LoRA not only counters this degradation but also achieves a higher Comp score than NegCLIP.

Image-Text Pair	Pre-trained CLIP	DAC-LLM	FSC-CLIP (Ours)
	Top-1: Two bowls of oranges are sitting on a metal surface.	☒ A table with some oranges and some cups.	☑ A table with some oranges and some apples.
	Top-2: A glass bowl filled with oranges on a table.	some oranges that are sitting on some wood	☒ A table with some oranges and some cups.
GT: ☑ A table with some oranges and some apples.	**Top-3:** ☒ A table with some oranges and some cups.	A glass bowl filled with oranges on a table.	A glass bowl filled with oranges on a table.
	Top-1: ☒ A man bending over a cake with a lot of candles.	a person holding a small cake with candles	☑ A man bending over a table with a lot of candles.
	Top-2: ☑ A man bending over a table with a lot of candles.	A person that is putting candles on a cake.	☒ A man bending over a cake with a lot of candles.
GT: ☑ A man bending over a table with a lot of candles.	**Top-3:** A man presents a cake with lit candles on it to a seated man.	A man blowing out candles on a cake	A man is being handed a birthday cake with lit candles.

Figure 5: Image to text retrieval examples on COCO-CF dataset. CLIP and DAC-LLM often rank negative captions (marked with red crossmarks) as top-1, while `FSC-CLIP` consistently retrieves the correct caption (marked with green checkmarks), demonstrating superior fine-grained understanding and retrieval accuracy in challenging conditions.

Qualitative Counterfactual Image to Text Retrieval Results. In Fig. 5, we compare image to text retrieval results on the COCO-Counterfactuals (COCO-CF) (Le et al., 2023) dataset for three models: pre-trained CLIP (Radford et al., 2021), DAC-LLM (Doveh et al., 2023), and our proposed `FSC-CLIP`. The figure displays the top-3 retrieved captions for each image, with correct captions indicated by green checkmarks and incorrect ones by red crossmarks. We observe that CLIP and DAC-LLM often fail to retrieve the correct caption associated with the image, ranking a negative caption as top-1. In contrast, our `FSC-CLIP` consistently retrieves the correct caption as top-1, demonstrating superior retrieval capabilities along with a stronger fine-grained compositional understanding, even in the presence of hard negative captions.

5 Conclusion

In this paper, we introduce Fine-grained Selective Calibrated CLIP (`FSC-CLIP`), a new fine-tuning framework for vision-language compositionality. It aims to preserve multi-modal capabilities and address the limitations of existing methods relying on global representations. We achieve this by employing dense representations between images and texts and regularizing the hard negative losses to prevent degradation, thereby facilitating the introduction of Local Hard Negative Loss and Selective Calibrated Regularization. Our extensive validation shows improved compositional reasoning and promising performance in standard multi-modal tasks.

Limitations. Our methodology, including all the prior approaches, relies on short captions for both training and evaluation benchmarks. This practice constrains the models' exposure to and understanding of longer contexts, which are essential for achieving a genuine vision-language compositional understanding. Longer and detailed captions involve more complex associations and contextual nuances (Onoe et al., 2024; Garg et al., 2024) that are essential for advanced compositionality in vision and language models. Moving forward, there is a compelling need within the community to develop training and evaluation protocols that incorporate longer captions, better addressing the challenges of compositionality.

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A Additional Details

A.1 Rule-based Hard Negative Texts

We provide details in generating hard negative texts in our model. We employ three types of rule-based methods: `negclip` (Yuksekgonul et al.,

Image-Text Pair	negclip	replace	bi-gram shuffle
 Three statues of elephants on the steps in front of an old building.	Three statues of steps on the elephants in front of an old building. Three statues of elephants on the steps in building of an old front. Three elephants of statues on the steps in front of an old building.	Three statues of pikas on the steps in front of an old building. Three statues of elephants into the steps in front of an old building. Three statues of megatherian mammal on the steps in front of an old building.	on the an old steps in building. Three statues front of of elephants Three statues building. of elephants an old steps in front of on the steps in on the front of an old building. Three statues of elephants
 Four different types of sandals with laces.	Four different sandals of types with laces. Four different laces of sandals with types. Four different types of laces with sandals.	Four different types of slingbacks with laces. Four inactive types of sandals with laces. Four different types of sandals with arms.	Four different laces. types of sandals with sandals with types of Four different laces. laces. types of Four different sandals with
 The small blue van is parked in front of a fence.	The blue small van is parked in front of a fence. The small blue van is parked in fence of a front.	The small blue regiment is parked in front of a fence. The small large van is parked in front of a fence. The small average van is parked in front of a fence.	is parked of a The small in front fence. blue van blue van in front of a is parked fence. The small The small in front blue van fence. is parked of a

Figure 6: Example results of rule-based hard negative texts used for training our model. Image-text pairs were randomly sampled from LAION-COCO (Schuhmann et al., 2022a). For `negclip` (Yuksekgonul et al., 2023) and `replace` (Hsieh et al., 2023), differences from the original captions are highlighted in red.

2023), `replace` (Hsieh et al., 2023), and `bi-gram shuffle`. Each method is implemented in an online version and applied to the original text at every training step, resulting in total of four texts including the original caption for every batch as illustrated in Fig. 2. In the online augmentation process, some captions do not yield a hard negative counterpart; these are masked out and excluded from the hard negative loss calculation.

The `negclip` method rearranges words within captions by swapping similar phrase types such as nouns, verbs, or adjectives within the text.

The `replace` method generates hard negative texts by replacing specific elements in the caption: entities, relations, or attributes, using antonyms or co-hyponyms from WordNet (Fellbaum, 2010).

The `bi-gram shuffle` rearranges text by shuffling bi-grams (*e.g.*, pairs of adjacent words), within a sentence. It varies the sentence structure, ensuring the generated texts serve as challenging negatives to the original.

All the augmentation methods above utilize the SpaCy (Honnibal and Montani, 2017) package. We implemented `bi-gram shuffle`, while for `negclip` and `replace`, we adopted the implementations from CLoVe (Castro et al., 2024). For illustrative purposes, we provide examples of each method applied to image-caption pairs, in Fig. 6.

A.2 Details on Evaluation Benchmark

Compositionality. VLMs are presented with either an image or text query and must identify the correct match from a set of candidates, which includes subtly altered incorrect options of texts and images. If there are two candidates, including the original, the random chance accuracy becomes 0.5.

Benchmarks are grouped into three categories based on the query modality. Tab. 6 provides a list of benchmarks for each category, along with the corresponding dataset licenses.

(1) `Image-to-Text`, where the objective is to choose the correct textual description for a presented image: ARO (Yuksekgonul et al., 2023), CREPE-Productivity (Ma et al., 2023), Sugar-Crepe (Hsieh et al., 2023), VALSE (Parcalabescu et al., 2022), VL-Checklist (Zhao et al., 2022), and WhatsUp (Kamath et al., 2023b).

(2) `Text-to-Image`, which requires the selection of the correct image that matches a given text query: ImageCoDE (Krojer et al., 2022) and SVO Probes (Hendricks and Nematzadeh, 2021).

(3) `Group`, which involves two counterfactual image-text pairs, the challenge is to match each image with its corresponding text and the vice versa: Winoground (Thrush et al., 2022), EqBen (Wang et al., 2023), and SPEC (Peng et al., 2024).

For the `Image-to-Text` and `Text-to-Image` tasks, top-1 accuracy is used. For the `Group` tasks, group accuracy measures whether VLMs correctly match all the associated image-text pairs.

Benchmark	License	Image source	Tasks and Subtasks
ARO	MIT	COCO, Visual Genome, Flickr30k	VG_Relation, VG_Attribution, Flickr30k_Order, COCO_Order
CREPE-Productivity	*unspecified*	Visual Genome	Atomic Foils, Negate, Swap
SugarCrepe	MIT	COCO	Add_{object, attribute}, Replace_{object, attribute, relation}, Swap_{object, attribute}
VALSE	MIT	Visual7w, COCO, SWiG, VisDial_v1.0, FOIL-it	Actions_{swap, replacement}, Coreference_{hard, standard}, Counting_{adversarial, hard, small}, Existence, Foil-it, Plurals, Relations
VL-Checklist	*unspecified*	Visual Genome, SWiG, COCO, HAKE, HICO_Det, Pic, HCVRD, OpenImages	Object_Location_{center, margin, mid}, Object_Size_{large, medium, small}, Attribute_{action, color, material, size, state}, Relation_{action, spatial}
WhatsUp	MIT	Controlled_Images (*self-captured*), COCO, GQA	Controlled_Images_{A, B}, COCO_QA_{One, Two}, VG_QA_{One, Two}
ImageCoDe	MIT	OpenImages, MSRVTT, Video-Storytelling, YouCook	Static (*e.g.*, images), Video (*e.g.*, videos)
SVO Probes	Apache-2.0	Google Image Search API	Subject, Verb, Object
Winoground	META IMAGES RESEARCH LICENSE	Getty Images	-
EqBen	Apache-2.0	Action Genome (AG), GEBC, YouCook2, Kubric, StableDiffusion (SD)	EQ-AG, EQ-GEBC, EQ-YouCook2, EQ-Kubric_{location, counting, attribute}, EQ-SD
SPEC	*unspecified*	Stable-Diffusion-XL 1.0	Absolute_size, Absolute_position, Count, Relative_size, Relative_position, Existence

Table 6: A comprehensive list of compositionality benchmarks used in our work, further subdivided based on the query types for each individual test: `Image-to-Text`, `Text-to-Image`, and Group, respectively.

To elaborate on details in specific benchmarks, for EqBen, we cap the evaluation sample size at 20,000. This is because the sub-tasks `eqbenag` and `eqbenyoucook2` contain 195,872 and 45,849 samples respectively, and evaluating all samples would be excessively time-consuming. Limiting the number of samples does not significantly alter the evaluation results. We do not use the official repository's 10% evaluation split because it does not support sub-task-specific evaluations.

For SVO-Probes, we have downloaded image-text pairs using the img2dataset (Beaumont, 2021) package from the URL list[1], as they are not available as physical files. Out of the original 36.8k samples, 22,162 were successfully downloaded, with 3,728 for the `subj_neg`, 13,523 for the `verb_neg`, and 4,911 for the `obj_neg` sub-tasks, respectively.

For SPEC, unlike the other datasets in the `Group` category, we use the average of image to text and text to image accuracy, rather than group accuracy.

Zero-shot Classification. We leverage ELEVATER toolkit (Li et al., 2022a) for 21 zero-shot classification tasks, including ImageNet (Deng et al., 2009), licensed under MIT License.

Image-Text Retrieval. We utilize COCO captions (Chen et al., 2015), Flickr30k (Young et al., 2014), and COCO-Counterfactuals (Le et al., 2023) to evaluate the retrieval task. These datasets are licensed under BSD-3-Clause, CC0: Public Domain, and CC-BY-4.0, respectively. For COCO-Counterfactuals, we randomly selected 30% of the total 17,410 samples for evaluation, resulting in 5,223 samples. Each example includes two counterfactual image-text pairs, so the total number of images and texts used for retrieval is 10,446; one for the original and one for the hard negatives.

A.3 Train Dataset

We used the pre-processed version of COCO captions (Chen et al., 2015) by Yuksekgonul et al. (2023), licensed under BSD 2-Clause. In addition, we utilized LAION-COCO (Schuhmann et al., 2022a), licensed under CC-BY-4.0, and CC-3M (Sharma et al., 2018)[2], with 100K randomly sampled examples from each dataset to match the size of COCO for fine-tuning. We downloaded both datasets using the img2dataset package.

[1] https://huggingface.co/datasets/MichiganNLP/svo_probes

[2] https://github.com/google-research-datasets/conceptual-captions/blob/master/LICENSE

id	Attn. Norm.	Comp	ZS	I2T Ret	T2I Ret
2	minmax	51.7	**55.7**	**61.6**	54.5
2	minmax-sparse	51.6	55.5	61.1	**54.8**
2	softmax	**52.0**	55.4	60.9	54.6
6	minmax	**53.5**	55.3	**58.2**	55.5
6	minmax-sparse	53.4	55.1	57.8	55.4
6	softmax	53.3	**55.5**	57.1	**55.7**

Table 7: Ablation study on the normalization of attention weights in Eq. (6) for the LHN Loss. We found that no specific normalization method significantly impacted the results, highlighting the importance of the unique LHN loss design.

A.4 Baseline Methods

In the comparisons with previous methods in Tab. 1, we evaluated prior approaches using the same protocol as ours to ensure fair and consistent evaluation. We obtained the corresponding checkpoints from each official repository and loaded them using the open_clip package (Ilharco et al., 2021).

When loading the checkpoints of previous models, we explicitly set `quick_gelu` to True in the open_clip implementation. While this setting was omitted in the implementations of NegCLIP (Yuksekgonul et al., 2023), CE-CLIP (Zhang et al., 2024), and GNM-CLIP (Sahin et al., 2024), the adjustment aligns with the original CLIP models from (Radford et al., 2021), which were pre-trained and also fine-tuned with this option activated.

We list the previous methods with corresponding licenses. NegCLIP (Yuksekgonul et al., 2023): MIT License, CE-CLIP (Zhang et al., 2024): MIT License, GNM-CLIP (Sahin et al., 2024): Apache-2.0 License, TSVLC[3] and DAC[4] (Doveh et al., 2022, 2023): *unspecified*, CLoVe (Castro et al., 2024): MIT License.

B Additional Results

For thoroughness, we include additional results not featured in the main paper. Note that all models were fine-tuned using the CLIP ViT-B/32 encoder from OpenAI (Radford et al., 2021).

B.1 Additional Analysis

Normalization of attention weights. We present an ablation experiment on the normalization of attention weights in Eq. (6), in alignment with the ablation study in Tab. 2. We replace the current `minmax` normalization with `minmax-sparse` (Bica et al., 2024) and `softmax`, respectively. As in Tab. 2, 'id 2' only applies the LHN Loss without global HN loss and SCR, while 'id 6' represents the full objective. Our findings show that the effectiveness of LHN Loss is not significantly impacted by any particular normalization technique. In other words, general normalization of attention weights can be applied to LHN Loss, reducing reliance on techniques like those from Bica et al. (2024). This suggests that the unique design of LHN Loss is key to the improved performance.

B.2 Multiple Runs

In Tab. 8, we report the mean and standard deviation for our models across all tasks listed in Tab. 1, using three distinct seeds: 0, 1, and 2 for training each model.

B.3 Zero-shot Classification

We report the results for each benchmark within the 21 zero-shot classification tasks in Tab. 9.

B.4 Image-Text Retrieval

We present the results for each benchmark included in the three image-text retrieval tasks in Tab. 10.

[3] https://github.com/SivanDoveh/TSVLC
[4] https://github.com/SivanDoveh/DAC

Method	LoRA	ARO	CREPE	EqBen	ImageCoDe	SugarCrepe	SVO Probes	VALSE	VL-Checklist	WhatsUp	Winoground	SPEC	Comp	ZS	I2T Ret	T2I Ret
Fine-tuned: LAION-COCO, 100K Samples																
FSC-CLIP		$82.7_{0.10}$	$46.6_{0.35}$	$29.3_{0.17}$	$24.6_{0.94}$	$82.6_{0.14}$	$90.1_{0.03}$	$73.5_{0.15}$	$75.7_{0.33}$	$42.1_{0.25}$	$6.2_{0.63}$	$33.5_{0.17}$	$53.4_{0.09}$	$55.6_{0.32}$	$57.8_{0.52}$	$55.3_{0.20}$
FSC-CLIP	✓	$85.3_{0.14}$	$52.9_{1.28}$	$28.9_{0.17}$	$24.9_{0.11}$	$80.5_{0.11}$	$89.7_{0.05}$	$72.4_{0.17}$	$78.7_{0.20}$	$42.9_{0.05}$	$5.4_{0.38}$	$32.4_{0.11}$	$54.0_{0.17}$	$56.1_{0.18}$	$57.3_{0.13}$	$54.4_{0.08}$

Table 8: Evaluation across three training runs of our model using different seeds. We report the mean and standard deviation obtained from the evaluation results of the models across three trials.

Method	caltech101	cifar10	cifar100	country211	dtd	eurosat-clip	fer2013	fgvc-aircraft-2013b	flower102	food101	gtsrb	hateful-memes	imagenet-1k	kitti-distance	mnist	oxford-iiit-pets	patchcamelyon	rendered-sst2	resisc45-clip	stanfordcar	voc2007classification	Average
CLIP-ViT-B/32	88.3	89.8	65.1	17.2	44.4	45.5	42.3	19.7	66.7	84.0	32.6	55.9	63.3	27.4	48.3	87.1	60.6	58.6	60.0	59.7	82.6	57.1
Fine-tuned: MS-COCO, 100K Samples																						
NegCLIP	88.2	88.9	63.2	15.0	43.1	47.3	47.6	16.8	62.3	79.4	30.2	54.3	60.9	27.6	49.7	85.4	59.7	58.8	56.9	54.0	84.4	55.9
CE-CLIP	82.2	85.9	60.2	9.6	35.2	44.9	39.7	10.0	47.2	70.1	28.0	53.5	49.9	34.6	40.6	66.0	58.8	61.1	51.5	35.3	83.1	49.9
GNM-CLIP	86.8	88.4	65.7	15.2	42.0	50.1	46.6	17.3	62.4	81.8	30.2	54.9	61.4	25.2	54.4	86.3	59.0	58.5	58.7	53.1	84.0	56.3
Fine-tuned: Conceptual Captions – 3M (CC-3M), 100K Samples																						
TSVLC (RB)	83.7	92.3	66.0	16.2	39.5	52.1	43.6	14.7	58.2	81.2	24.2	57.8	58.5	30.4	46.9	85.5	50.0	59.8	58.6	49.2	84.7	54.9
TSVLC (RB+LLM)	84.6	92.0	66.8	16.2	40.3	56.5	46.8	13.8	58.5	81.6	27.1	56.9	59.7	27.8	43.9	84.7	50.5	60.1	59.5	50.5	84.7	55.4
DAC-LLM	82.6	90.4	64.1	14.3	38.4	52.5	50.7	10.5	49.7	74.1	24.2	56.3	51.0	16.3	42.1	74.4	50.0	54.5	52.2	39.4	85.1	51.1
DAC-SAM	81.3	89.9	64.1	14.8	40.4	49.8	48.0	8.9	48.9	72.3	24.9	55.7	52.3	18.7	45.2	76.7	58.9	60.0	54.7	39.8	84.1	51.9
Fine-tuned: LAION-COCO, 600M Samples																						
CLoVe	85.5	85.8	66.2	12.6	37.7	49.1	38.0	9.0	44.6	71.9	22.6	54.6	53.1	34.9	36.4	74.2	56.7	51.3	55.2	48.7	81.9	51.0
Fine-tuned: LAION-COCO, 100K Samples																						
FSC-CLIP (Ours)	86.5	87.5	65.7	15.3	42.4	43.9	48.9	14.9	55.5	80.5	31.6	55.9	58.1	29.1	52.4	84.2	61.0	56.0	56.9	52.0	83.6	55.3
FSC-CLIP (Ours, LoRA)	85.9	88.5	66.3	15.8	39.8	52.8	48.2	14.2	57.0	81.0	27.9	56.3	57.4	33.9	54.3	82.7	59.8	57.2	58.7	52.6	83.7	55.9

Table 9: Expanded results for the 21 zero-shot classification tasks from ELEVATER (Li et al., 2022a).

	COCO Retrieval						Flickr30k Retrieval						COCO-Counterfactuals Retrieval						Avg.	
	Image to text (I2T)			Text to image (T2I)			Image to text (I2T)			Text to image (T2I)			Image to text (I2T)			Text to image (T2I)			I2T	T2I
Method	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@1
CLIP-ViT-B-32	50.1	74.9	83.5	30.4	56.0	66.8	78.8	94.9	98.3	58.7	83.5	90.0	51.0	79.3	86.7	48.1	77.4	85.9	60.0	45.8
Fine-tuned: MS-COCO, 100K Samples																				
NegCLIP	59.3	82.8	89.4	45.2	72.1	81.7	85.7	96.4	98.8	71.6	91.8	95.7	55.3	82.5	89.2	58.3	84.9	91.3	66.8	58.4
CE-CLIP	56.0	81.6	89.0	47.1	74.1	83.1	75.3	93.2	96.9	68.9	89.6	94.2	46.3	75.7	84.5	56.2	83.6	90.5	59.2	57.4
GNM-CLIP	58.1	81.4	88.8	41.1	67.5	77.8	82.9	96.2	98.6	68.8	89.9	94.1	57.2	84.5	90.5	56.7	84.5	91.1	66.1	55.5
Fine-tuned: Conceptual Captions – 3M (CC-3M), 100K Samples																				
TSVLC (RB)	46.1	71.7	80.4	36.3	62.0	72.4	74.0	93.2	96.4	64.9	87.2	92.7	44.6	72.0	80.2	55.0	83.3	90.0	54.9	52.1
TSVLC (RB+LLM)	46.4	71.8	80.8	36.6	62.2	72.7	74.8	92.6	96.8	65.1	87.6	92.7	44.1	71.5	80.1	55.1	83.3	90.4	55.1	52.3
DAC-LLM	29.9	54.5	65.6	37.3	63.5	73.8	52.9	79.8	87.9	64.6	88.0	93.0	28.1	53.6	64.4	55.2	83.0	90.0	36.9	52.4
DAC-SAM	33.1	57.9	68.8	34.0	59.7	70.0	59.8	82.7	89.0	61.7	85.7	91.2	30.4	55.2	64.8	51.5	79.9	87.3	41.1	49.0
Fine-tuned: LAION-COCO, 600M Samples																				
CLoVe	48.3	73.9	82.8	42.7	68.7	78.2	69.5	90.4	95.6	68.7	90.0	94.5	41.5	69.1	78.3	56.5	84.2	90.8	53.1	56.0
Fine-tuned: LAION-COCO, 100K Samples																				
FSC-CLIP (Ours)	49.7	73.6	82.4	40.4	66.4	76.4	75.6	93.3	97.4	68.2	90.0	94.3	49.2	77.5	85.8	57.9	85.4	91.4	58.2	55.5
FSC-CLIP (Ours, LoRA)	48.2	73.6	81.8	39.0	64.9	75.0	75.1	93.2	96.4	66.9	88.6	93.6	48.5	76.0	84.4	57.1	84.7	91.0	57.3	54.3

Table 10: Expanded results for the three zero-shot image-text retrieval tasks, including COCO (Chen et al., 2015), Flickr30k (Young et al., 2014), and COCO-Counterfactuals (Le et al., 2023).